

DD_OSR-02 - Procurement - Developer Implementation Guide

Version: v1.0 **Bundle:** 08_osr_equipment **Companion DD:** DD_OSR-02-Procurement-v1.0

1. Goal

Implement the procurement flow that creates approved purchase orders and posts vendor deliveries into OSR stock.

The key implementation rule is ownership:

- OSR-02 owns PO, goods receipt, and procurement discrepancy records.
- OSR-01 owns stock movements and balances.
- OSR-INSTANCE-01 owns serialized device rows.
- EM-CFG-04 owns approval tasks and approval process configuration.

2. Runtime Flow

```
sequenceDiagram
    participant B0 as Backoffice
    participant P0 as OSR-02
    participant PLM as PLM-CFG-06
    participant APR as EM-CFG-04
    participant OSR as OSR-01
    participant INST as OSR-INSTANCE-01

    B0->>P0: POST /api/procurement/orders
    P0->>PLM: validate SKU
    B0->>P0: POST /submit-approval
    P0->>APR: evaluate/start approval
    APR-->>P0: approval callback
    B0->>P0: POST /api/procurement/goods-receipts
    P0->>OSR: POST /api/stock-movements
    P0->>INST: POST /api/equipment-instances/bulk-import
    P0-->>B0: receipt posted
```

3. Step 1 - Migrations

Create:

- procurement_vendor
- procurement_order
- procurement_order_line
- goods_receipt
- goods_receipt_line
- procurement_discrepancy

Add comments in migrations so a developer can inspect DB metadata and understand purpose.

Minimum uniqueness:

- (operator_code, vendor_code) on vendor.
- (operator_code, po_number) on PO.
- (operator_code, vendor_delivery_ref) on receipt.

4. Step 2 - Create PO

Endpoint:

```
POST /api/procurement/orders
Idempotency-Key: po-wik-huawei-20260715
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
  "vendorId": "ven-huawei-ke",
  "destinationLocationId": "WIK-CENTRAL-NRB",
  "currencyCode": "KES",
  "expectedDeliveryDate": "2026-07-15",
  "lines": [
    {
      "skuId": "WIK-ONT-HUAWEI-EG8145V5",
      "orderedQty": 200,
      "unitCost": 6000.00
    }
  ]
}
```

On OSR-02:

1. Validate request and idempotency key.
2. Load vendor; must be ACTIVE.
3. Load destination location from OSR-01 GET /api/stock-locations/{location_id}.
4. For each line, load SKU from PLM-CFG-06 and copy isSerialized.
5. Compute totals.
6. Create procurement_order in DRAFT.
7. Create procurement_order_line rows.
8. Emit ProcurementOrderCreated.

Do not call OSR-01 stock movement during PO creation.

5. Step 3 - Submit PO For Approval

Endpoint:

```
POST /api/procurement/orders/po-2026-00045/submit-approval
```

Flow:

1. Load PO and lines.
2. Run fixed validation.
3. Build Drools fact.
4. Run procurement-order-approval.drl.
5. Call EM-CFG-04 POST /api/approvals/evaluate.
6. If approval not required, set PO APPROVED.
7. If approval required, call POST /api/approvals/requests.
8. Store approval_request_id, approval_mode, camunda_process_instance_id.
9. Set PO PENDING_APPROVAL.

EM-CFG-04 evaluate request:

```
{
  "operatorCode": "WIK",
  "moduleCode": "OSR-02",
  "actionCode": "PROCUREMENT_ORDER_APPROVAL",
  "subjectType": "PROCUREMENT_ORDER",
  "subjectId": "po-2026-00045",
  "context": {
    "totalEstimatedAmount": 1200000.00,
    "currencyCode": "KES",
  }
}
```

```

        "vendorId": "ven-huawei-ke",
        "lineCount": 1
    }
}

```

Camunda note: OSR-02 does not start Camunda. EM-CFG-04 starts Camunda only if the selected approval policy is BPMN.

6. Step 4 - Approval Callback

Endpoint:

```

POST /internal/procurement/orders/po-2026-00045/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-po-00045",
  "outcome": "APPROVED",
  "decisionComment": "Approved for Nairobi stock replenishment."
}

```

Implement idempotently:

1. Load PO.
2. Verify request id matches PO.
3. If already processed, return 200.
4. APPROVED -> PO status APPROVED.
5. REJECTED -> PO status REJECTED.
6. Emit event.

7. Step 5 - Record Goods Receipt

Endpoint:

```

POST /api/procurement/goods-receipts
Idempotency-Key: gr-dn-hw-7788
Content-Type: application/json

```

Flow:

1. Validate request and idempotency key.
2. Load PO; status must be APPROVED or SENT_TO_VENDOR.
3. Validate vendor delivery ref uniqueness.
4. For each line, match poLineId and skuId.
5. Validate quantities: acceptedQty + rejectedQty = receivedQty.
6. For serialized SKUs, validate serial count equals accepted qty.
7. Detect duplicates inside payload.
8. Optionally call OSR-INSTANCE lookup by serial to catch existing records.
9. Create goods_receipt and goods_receipt_line in VALIDATED.
10. Create procurement_discrepancy rows where needed.
11. Post accepted quantities to OSR-01.
12. Import accepted serialized units to OSR-INSTANCE-01.
13. Mark receipt POSTED.
14. Update PO line received qty and PO status.
15. Emit GoodsReceiptPosted.

8. Step 6 - Call OSR-01 Stock Movement

For accepted quantity:

```
POST /api/stock-movements
Idempotency-Key: stock-move-gr-001-grl-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "movementType": "INBOUND",
  "reasonCode": "VENDOR_RECEIPT",
  "skuId": "WIK-ONT-HUAWEI-EG8145V5",
  "quantity": 148,
  "targetLocationId": "WIK-CENTRAL-NRB",
  "originRefType": "GOODS_RECEIPT",
  "originRefId": "gr-001",
  "actorUserId": "usr-warehouse-001"
}
```

If OSR-01 fails, keep receipt not posted and retry safely. Do not create serial instances without matching quantity movement unless you have a compensation/retry state.

9. Step 7 - Call OSR-INSTANCE For Serialized SKUs

For accepted serialized units:

```
POST /api/equipment-instances/bulk-import
Idempotency-Key: instance-import-gr-001-grl-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "skuId": "WIK-ONT-HUAWEI-EG8145V5",
  "initialState": "IN_MAIN_WAREHOUSE",
  "initialLocationId": "WIK-CENTRAL-NRB",
  "originRefKind": "GOODS_RECEIPT",
  "originRefId": "gr-001",
  "instances": [
    {
      "serialNumber": "48575443ABCD1234",
      "macAddress": "00:11:22:33:44:55"
    }
  ]
}
```

If import partially fails because of duplicate serials:

- do not silently post partial success;
- create procurement_discrepancy with DUPLICATE_SERIAL;
- require warehouse/procurement resolution.

10. Step 8 - Discrepancy Resolution

Endpoint:

```
POST /api/procurement/discrepancies/pd-001/resolve
Content-Type: application/json
```

Flow:

1. Load discrepancy.
2. Run goods-receipt-discrepancy.drl.
3. If approval required, call EM-CFG-04 using action PROCUREMENT_DISCREPANCY_RESOLUTION.
4. If approved/not required, apply resolution:
 - ACCEPT: post remaining accepted stock if valid.
 - REJECT_TO_VENDOR: leave quantity out of stock and close discrepancy.
 - CREDIT_NOTE: record resolution for finance/reporting.
 - REPLACEMENT_PENDING: keep discrepancy open with expected vendor replacement.

5. Emit ProcurementDiscrepancyResolved.

11. Drools Samples

```
package com.sophix.osr.procurement

rule "High value PO requires finance approval"
when
    $po : ProcurementOrderFact(totalAmount >= 1000000.00, currencyCode == "KES")
    $d : ProcurementDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("PROCUREMENT_HIGH_VALUE_FINANCE");
end

rule "Damaged receipt above five units requires approval"
when
    $disc : ProcurementDiscrepancyFact(discrepancyType == "DAMAGED", quantity > 5)
    $d : ProcurementDecision()
then
    $d.setApprovalRequired(true);
    $d.setApprovalPolicyCode("PROCUREMENT_DAMAGED_RECEIPT_REVIEW");
end
```

12. Worker Jobs

These are command/job handlers in the OSR worker deployment:

Job	Purpose
procurement-outbox-publish	Publish PO/receipt/discrepancy events.
goods-receipt-posting-retry	Retry failed OSR-01/OSR-INSTANCE posting.
vendor-delivery-reminder	Notify procurement for overdue deliveries.
procurement-discrepancy-reminder	Notify owners for unresolved discrepancies.

13. Tests

- Create PO with active vendor and active SKU.
- Reject PO with inactive vendor.
- Submit approval and store EM-CFG-04 request.
- Approval callback is idempotent.
- Goods receipt posts OSR-01 movement.
- Serialized goods receipt calls OSR-INSTANCE bulk import.
- Duplicate vendor delivery ref does not double post stock.
- Duplicate serial creates discrepancy.
- Discrepancy resolution calls EM-CFG-04 when configured.